

# MODEL LESSON — TEACHER SCRIPT & COACHING GUIDE

## Factoring Trinomials $x^2 + bx + c$ ( $a = 1$ ) • The X-Puzzle • TEKS A.10E

Waltrip High School • Algebra 1 • Vigneshwar Lokoji • Designed for 15–20 min (core 17)

### 1. At a Glance

#### Learning Objective (LO):

I can factor a trinomial  $x^2 + bx + c$  ( $a = 1$ ) into two binomials.

#### Demonstration of Learning (DOL):

Given 4 trinomials, students factor at least 3 of 4 correctly on the exit ticket.

#### Essential Question (EQ):

How do the  $b$  and  $c$  terms of a trinomial help us factor it?

#### The big idea I keep stamping:

**$c$  is the PRODUCT.  $b$  is the SUM.** Find the two numbers, then fix the signs.

#### One-line framing for the panel:

*"This lesson is built on the Gradual Release model — I Do, We Do, You Do. My job is to hand the thinking to students on purpose: I model briefly, we practice with peer support, and then they prove it independently. Everything is color-coded so you can see the release happen."*

### Timing & SPOT Look-Fors

| Slide                 | Phase                                                                                           | Time | Primary SPOT / "Rock the Boat" look-for              |
|-----------------------|-------------------------------------------------------------------------------------------------|------|------------------------------------------------------|
| 1–2 Title + Flow      | Frame                                                                                           | 0:45 | Release model made VISIBLE & intentional             |
| 3 Do Now              | Warm-up                                                                                         | 1:45 | At-bat #1; whiteboards = check for understanding     |
| 4 Objective + DOL     | Frame                                                                                           | 0:35 | LO/DOL posted & referenced; teach to the DOL         |
| 5 What do you notice? | Connect                                                                                         | 1:15 | Annotation; use student answers; stamp               |
| 6 I Do                |  Model       | 2:45 | Model THINKING not completion; name decisions; brief |
| 7 Sign chart          | Scaffold                                                                                        | 0:40 | Intentional scaffold that will EXPIRE                |
| 8 We Do #1            |  Guided      | 1:45 | Peer support; purposeful turn & talk; stamp          |
| 9 We Do #2            |  Guided      | 2:15 | Monitor misconceptions (signs), don't re-model       |
| 10 You Do             |  Independent | 2:45 | Silent, timed, every student commits, NO rescue      |
| 11 Exit Ticket (DOL)  |  Assess      | 1:30 | The DOL; monitor & adjust from data                  |
| 12 Debrief + Stamp    | Close                                                                                           | 0:45 | Engagement (speaking); stamp the key point           |
| 13 Close              | Close                                                                                           | 0:15 | Thank the panel                                      |

**Flex:** Running long? Do one We Do and demo 2 exit-ticket items. Have time? Extend We Do #2 and add a perfect-square trinomial as an extension.

## Materials & Non-Negotiables I'm Hitting

### On each student desk:

- Whiteboard + marker (for Do Now, We Do, You Do — writing every few minutes)
- 3 response cards (A/B/C/D) for quick multiple-choice checks
- Student Response Handout (printed)

### I will, on purpose:

- Make the release model visible (Slide 2) and name it
- Require writing & thinking before talking
- State a PURPOSE before every turn & talk
- Circulate constantly — including the right-hand side of the room
- Protect the independent “You Do” — no rescue

## 2. Word-for-Word Script (slide by slide)

“SAY” = say it close to verbatim. “DO” = your moves. “CHECK” = the check for understanding. “WATCH” = likely misconception + fix. “SPOT” = rubric credit.

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### Slide 1 • Title [Frame] (0:20)

**SAY** “Good morning, mathematicians — welcome! I’m Mr. Lokoji. Today we become factoring detectives: we take a trinomial apart, back into the two binomials it came from.”

**SAY** (Point to the three pills.) “Here’s the deal today: I DO — I model my thinking. WE DO — we practice together. YOU DO — you prove it on your own.”

**DO** Warm, high energy. Make eye contact. Move to the front-center.

**SPOT** Sets up the Gradual Release model out loud (planning + culture).

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### Slide 2 • Today’s Flow [Frame] (0:25)

**SAY** “Watch the arrow at the bottom. Right now I hold all the responsibility. By the end, YOU hold it. My whole job is to hand the work over to you — step by step.”

**DO** Tap each card: I Do → We Do → You Do. Keep it to ~20 seconds; do not teach math here.

**SPOT** Release model VISIBLE & intentional (a “Rock the Boat” look-for).

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### Slide 3 • Do Now [Warm-up] (1:45)

**SAY** “On your whiteboard, multiply each pair of binomials. You’ve got 2 minutes — whiteboards UP when you’re done. Use FOIL or the box method.”

**DO** Start the timer. Circulate immediately — scan the right side too. Pick 1–2 students to share on the next slide.

**CHECK** “Whiteboards UP.” That’s your live data on who remembers FOIL.

**WATCH** Forgetting to combine middle terms → “You have two x-terms — combine them into one.”

**SPOT** At-bat #1; engagement (writing); monitor; exemplar of success.

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### Slide 4 • Today’s Objective & DOL [Frame] (0:35)

**SAY** “Read the objective with me... ‘I can factor a trinomial into two binomials.’ Factoring is just UN-multiplying — exactly the reverse of what you did in the Do Now.”

**SAY** “And here’s your target: by the end you’ll factor 4 of these on your own — aim for at least 3 of 4.”

**DO** Have students read the LO chorally. Point to the word “factor.” Reference this DOL again at You Do and Exit Ticket.

**SPOT** Teach at grade level; teach to the DOL; thread objective throughout.

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#### Slide 5 • What Do You Notice? [Connect] (1:15)

**SAY** “Look back at your Do Now. There’s a hidden pattern. Turn and talk — 30 seconds: by the end you should be able to tell me how the 3 and the 5 are hiding inside  $x^2 + 8x + 15$ .”

**SAY** (After) “So what did you find? ... Right — the two numbers ADD to the middle and MULTIPLY to the last. Factoring just runs that backwards.”

**DO** State the turn-&-talk PURPOSE first. Then cold-call 2 partners. Annotate LIVE: circle 3 & 5, arrow to the 8 (sum) and 15 (product). Repeat on the negative example.

**CHECK** Cold-call after the turn & talk; listen for “add” and “multiply.”

**WATCH** Mixing up which is sum vs product → anchor with the annotation arrows.

**SPOT** Annotation; engagement (listening/speaking); use student answers; stamp.

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#### Slide 6 • I Do — Factor $x^2 + 7x + 12$ [I DO (model)] (2:45)

**SAY** “Watch how I DECIDE, not just what I write.”

**SAY** “First decision: is there a GCF? I look at 1, 7, 12 — no common factor. No GCF. Done.”

**SAY** “Now the heart of it: I need two numbers that MULTIPLY to my c, +12, and ADD to my b, +7. I’ll use the X-puzzle — product on top, sum on the bottom.”

**SAY** “Signs: product is positive and sum is positive, so both factors are positive.”

**SAY** “Test pairs: 1 and 12 add to 13 — no. 2 and 6 add to 8 — no. 3 and 4 add to 7 — yes!”

**SAY** “So my factors are  $(x + 3)(x + 4)$ . And I ALWAYS check by multiplying back:  $x^2 + 7x + 12$ . It matches.”

**DO** Pencils down, eyes up while you model. Fill the X-puzzle as you talk. Keep it under 3 minutes — resist solving a second one.

**CHECK** “Thumbs up if you could follow each decision.”

**WATCH** Over-talking. The #1 drift risk: model ONCE, then release.

**SPOT** Model THINKING not completion; name decisions; brief; exemplar; stamp.

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#### Slide 7 • Sign Chart (Scaffold) [Scaffold] (0:40)

**SAY** “Quick tool. Same signs? Both match c’s neighbor. Different signs — that’s when c is negative — the bigger number wins the sign of b.”

**SAY** “Use this on the next two problems. By the time you’re on your own, I want it in your HEAD, not on the screen.”

**DO** Walk the four cases fast with the class chiming in. Don’t lecture line by line.

**CHECK** Quick choral: “c positive, b negative — what are the signs?” (both negative)

**WATCH** Sign errors are the lesson’s #1 misconception — this chart pre-empts them.

**SPOT** Scaffolds (intentional, aligned) — and you SAY it will expire.

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#### Slide 8 • We Do #1 — Factor $x^2 + 9x + 20$ [WE DO (guided)] (1:45)

**SAY** “We’ll fill the X-puzzle together — you call the numbers. GCF? ... None. What multiplies to 20 and adds to 9?”

**SAY** (Cold-call) “Marcus — give me a pair... 4 and 5, nice. Signs? Both positive — why?”

**SAY** “Turn & talk, 30 seconds: convince your partner WHY both factors are positive, using the sign chart.”

**DO** No-opt-out: have students supply each number. Fill the puzzle live. State the turn-&-talk purpose first, then circulate and pick a pair to share.

**CHECK** Cold-call for each blank; listen during turn & talk for sign reasoning.

**WATCH** Don’t re-model the whole thing. If a pair is stuck, ask a question — don’t give the answer.

**SPOT** Peer support first; purposeful turn & talk; use student answers; stamp.

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#### Slide 9 • We Do #2 — Factor $x^2 + 2x - 15$ [WE DO (guided)] (2:15)

**SAY** “Tougher one — the  $c$  is NEGATIVE. What must be true about the signs?” (one +, one -)

**SAY** “ $b$  is +2 — so which one is positive, the bigger or the smaller? Convince me.”

**SAY** “Turn & talk: which factor gets the + sign, and how do you KNOW? Defend it to your partner.”

**DO** Lead with the QUESTION, not the answer. Monitor misconceptions during the turn & talk. Use a student’s correct justification to stamp it.

**CHECK** Verbal check:  $5 \cdot (-3) = -15$  and  $5 + (-3) = +2$ . ✓

**WATCH** Right pair, wrong signs (e.g., -5 and +3). Re-ask — don’t re-teach from scratch.

**SPOT** Monitor misconceptions not completion; questioning; use student answers.

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#### Slide 10 • You Do — Independent [YOU DO (independent)] (2:45)

**SAY** “Your turn to row the boat — I’m not rescuing this one. Silent. 3 minutes. Show your X-puzzle. Everyone commits to an answer on the whiteboard.”

**SAY** (After timer) “Whiteboards UP! ... Now confirm with your partner — to confirm, not to create.”

**DO** Start the timer, then GO QUIET. Circulate and MONITOR (right side too) — collect data with your eyes. NO rescue: one question max, then move on.

**CHECK** “Whiteboards UP” = live mastery data. THEN partners confirm (talking after thinking).

**WATCH** #2 sign errors. If many miss it, name it as tomorrow’s focus — out loud, for the observers.

**SPOT** Protect silence; timed; every student commits; no rescue; monitor & adjust.

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#### Slide 11 • Exit Ticket — DOL [DOL (assess)] (1:30)

**SAY** “This is your Demonstration of Learning — what I collect. Factor all four. Name on top. Hand it to me on your way out.”

**DO** In the demo: have the panel do #1 and #3, then narrate how you’d grade and use the rest.

**CHECK** Sort into mastery (3–4 of 4) vs not. The 4 items map to the 4 sign cases.

**WATCH** If a student misses only #4, you know it’s the negative-sum case — that’s your reteach.

**SPOT** Teach to the DOL; monitor & adjust; engagement (writing).

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#### Slide 12 • Debrief & Stamp [Close] (0:45)

**SAY** “Back to our Essential Question. Finish the frame: ‘I factor by finding two numbers that \_\_\_ to  $c$  and \_\_\_ to  $b$ .’”

**SAY** “Say the stamp with me:  $c$  is the PRODUCT,  $b$  is the SUM. That’s exactly what your exit ticket showed.”

**DO** Cold-call 2–3 students for the sentence frame. Everyone says the stamp aloud. Connect back to the LO/DOL.

**CHECK** Sentence-frame completion (speaking).

**SPOT** *Engagement (speaking); stamp; thread objective start-to-finish.*

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**Slide 13 • Close** [Close] (0:15)

**SAY** *“Thank you — I’d love to bring this kind of student-owned thinking to Ram Nation every day. I’m happy to take any questions.”*

**DO** Smile. Thank the panel by name if you can. Invite questions.

**SPOT** *Classroom culture; professionalism.*

### 3. Answer Keys

**Do Now:**  $(x + 3)(x + 5) = x^2 + 8x + 15$        $(x + 6)(x - 2) = x^2 + 4x - 12$

**I Do:**  $x^2 + 7x + 12 = (x + 3)(x + 4)$

**We Do #1:**  $x^2 + 9x + 20 = (x + 4)(x + 5)$

**We Do #2:**  $x^2 + 2x - 15 = (x + 5)(x - 3)$

**You Do:**  $x^2 - 7x + 12 = (x - 3)(x - 4)$        $x^2 + 3x - 10 = (x + 5)(x - 2)$

**Exit Ticket:** 1)  $(x + 3)(x + 5)$  2)  $(x - 4)(x - 5)$  3)  $(x + 4)(x - 3)$  4)  $(x - 7)(x + 3)$

**Why these four exit items:** #1 both +, #2 both -, #3 mixed (sum +), #4 mixed (sum -). Each tests a different sign case, so a wrong answer tells you exactly what to reteach.

### 4. Anticipated Misconceptions & Your Response

| Misconception                                  | Your move                                                                   |
|------------------------------------------------|-----------------------------------------------------------------------------|
| Wrong signs (right number pair) — the #1 error | "What's the sign of c? Of b?" Re-anchor to the sign chart — don't re-solve. |
| Adds instead of multiplies (or vice versa)     | "Top of the X is the product, c. Bottom is the sum, b."                     |
| Skips the GCF check                            | "Step 1 is always: any GCF?" (Build the habit even when there isn't one.)   |
| Doesn't verify the answer                      | Model FOIL-back every time: "Multiply it out — does it match?"              |
| Negative c → picks two same-sign numbers       | "If c is negative, the signs must be DIFFERENT — one +, one -."             |
| Stuck staring (no attempt)                     | "List the factor pairs of c first." One nudge, then move on — no rescue.    |

### 5. Questions the Panel May Ask (with strong answers)

**Q:** How do you differentiate for struggling students?

**A:** Layered scaffolds: the X-puzzle visual, the sign chart, sentence stems, and an X-puzzle template on the handout. Peers support in We Do. Early finishers get an extension (a perfect-square trinomial, or one with a leading coefficient to factor out first).

**Q:** How do you support Emergent Bilinguals (ELPS)?

**A:** ELPS 2I — listening comprehension — is built in. I pre-teach vocabulary (factor, trinomial, binomial, product, sum, coefficient), use the visual X-puzzle, provide sentence frames ("I factor by..."), color-code, and use structured partner talk so they rehearse language before writing.

**Q:** What are your checks for understanding?

**A:** Whiteboards-up (Do Now & You Do), no-opt-out cold calls (We Do), purpose-driven turn & talks, response cards for multiple-choice items, and the exit-ticket DOL sorted 3-of-4.

**Q:** How would you use the exit-ticket data?

**A:** Sort into mastery vs not. Because the 4 items map to the 4 sign cases, the specific item missed tells me the exact reteach. Those students form tomorrow's small group while the rest start an extension.

**Q:** What if a student finishes early?

**A:** Extension on the handout: a perfect-square trinomial ( $x^2 - 24x + 144$ ) or one needing a GCF first ( $2y^2 - 18y - 72$ ). Or: "Write the rule you used in your own words," or coach a partner.

**Q:** How do you handle a wrong answer?

**A:** I use it. “Walk me through your thinking” → find the decision point → ask a targeted question → let peers weigh in. I never just say “no.” Errors are data and teaching moments.

**Q: Why the X-puzzle method?**

**A:** It’s visual and low-floor/high-ceiling. It separates the two conditions (product on top, sum on bottom) so students manage one idea at a time, and it transfers directly to harder cases ( $a \neq 1$ ) later.

**Q: Is this on grade level / rigorous?**

**A:** Yes — TEKS A.10E. Rigor comes from multiple sign cases, justification (“how do you KNOW?”), and independent retrieval under time, not from harder arithmetic.

**Q: What’s your classroom-management plan?**

**A:** Tight, timed transitions; the “whiteboards up” routine; constant circulation with intentional proximity (incl. the right side); reinforce positives by name and redirect quietly without breaking momentum.

## 6. Delivery Tips — How to Score High (SPOT + “Rock the Boat”)

**Do:**

- Model THINKING, name decisions, and keep the I Do under ~3 minutes.
- State the PURPOSE before every turn & talk (“By the end of 30 sec you should be able to…”).
- In We Do, ask questions instead of re-modeling; let peers carry the work.
- In You Do, protect the silence, start the timer, circulate, and withhold rescue.
- Say out loud that scaffolds will expire — then remove them.
- Stamp the key point repeatedly: “c is the product, b is the sum.”
- Thread the objective from start to finish and reference the DOL.
- Narrate how you’d use the exit-ticket data (monitor & adjust).

**Don’t:**

- Over-talk during the model or solve more than one example.
- Re-teach the whole problem when a student is stuck — ask one question.
- Let discussion happen before independent effort in the You Do.
- Rescue too early — that robs students of the proof they can do it alone.
- Teach only from the front — keep moving.

## 7. 60-Second Pre-Flight Checklist

- ☐ LO & DOL posted and visible
- ☐ Whiteboards + markers + 3 response cards on each desk
- ☐ Student handout distributed
- ☐ Timer ready (2:00, 3:00)
- ☐ Answer keys memorized cold
- ☐ First 60 seconds rehearsed
- ☐ Energy on: smile, use names, move with purpose